SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMKUR

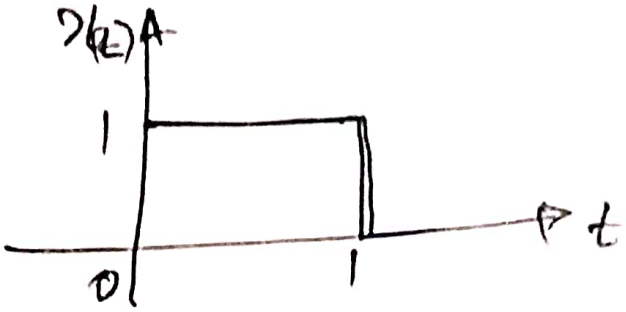
(A constituent college of Sri Siddhartha Academy of Higher Education, Tumakuru)

22EC305: SIGNALS AND SYSTEMS

Max. Marks:30 TEST1 Time:1.00Hr

Answer all the questions

		M	C	B
1	What is a Signal? Illustrate with examples, with neat diagrams and equations.	4	1	1
2	List and discuss the classifications of Signals, with neat diagrams and equations.	6	1	2
3	Explain the following properties of System: i. Causality ii. Time Invariance	4	2	3
4	Sketch the Signal $x_{(t)}=1-\|t\|$ for $-1\leq t\leq 1$ and derive the following Signals: **i.** $x_{(3t)}$ **ii.** $x_{(3t)} + x_{(3t-2)}$	6	2	3
5	Sketch the even and Odd component of the following Signal	4	2	3



6.	Sketch the Convolution Sum of the Following Signals	6	2	3

$$y(n) = u(n) * u(n-3)$$

NOTE: M is marks, C is CO and B is Blooms level

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Max. Marks:30 **TEST2** Time:1.00Hr

Answer all the questions

		M	C	B
1	State and Prove all the properties of Discrete Time Fourier Transform (DTFT).	4	2	2
2	Compute the forced response of the system given by the difference equation: $y(n)-(1/4)y(n-1)-(1/8)y(n-2)=x(n)+x(n-1)$ with input $x(n)=(1/8)^n u(n)$.	6	2	2
3	Compute the forced response for the system given by the differential equation, $\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=x(t)+\frac{dx(t)}{dt}$, With input $x(t)=5u(t)$. with zero initial condition.	6	2	3
4	Obtain the block diagram representations: Direct form-I and Direct form-II for a system modeled by the equation: $y(n)+(1/2)y(n-1)-y(n-3)=3x(n-1)+2x(n-2)$			

NOTE: M is marks, C is CO and B is Blooms level

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22EC305: Signals and Systems

Max Marks:20 **TEST3** Time:1.00Hr

Answer all the questions

		M	C	B
1	Determine the Z-Transform of the sequence: $x(n)=-u(-n-1)+(0.5)^n u(n)$. Depict the ROC and the locations of Poles and zeros of X(z) in the Z-Plane.	4	2	2
2	Discuss the properties of Z-Transform	6	3	2
3	Obtain the inverse Z-transform of the following signal: $X(z)=\frac{(0.25)z^{-1}}{(1-(0.5)z^{-1})(1-(0.25)z^{-1})}$ with ROC $0.25<\|z\|<0.5$	4	2	2
4	Solve the following difference equation $y(n)+3y(n-1)=x(n)$, with $x(n)=u(n)$ and the initial condition $y(-1)=1$.	6	4	3

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SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY- TUMAKURU

(A constituent College of Siddhartha Academy of Higher Education, Tumakuru)

B.E., SEMESTER END EXAMINATION JAN - FEB 2024

22EC305: SIGNALS AND SYSTEMS

TIME: 3.00 Hrs. **SEM: III** **MAX MARKS: 100**

NOTE: Answer any five full questions selecting one full question from each choice.

		M	CO	BL
1.a)	Define a signal and a system. Illustrates with examples.	6	1	2
b)	Check whether the following signals are periodic or not. If periodic, determine their fundamental period.	6	2	2

i. $x(n) = \cos\left(\frac{\pi n}{7}\right)\sin\left(\frac{\pi n}{3}\right)$ ii. $x(t) = \left[2\cos^2\left(\frac{\pi t}{2}\right) - 1\right]\sin \pi t \cos \pi t$

iii. $x(n) = \cos\left(\frac{2\pi n}{5}\right) + \cos\left(\frac{2\pi n}{7}\right)$

c)	List and discuss the classifications of Signals, with neat diagrams and equations.	8	1	2

OR

2.a)	Sketch the even and odd component of the following signal x(t)=1 for 0≤t≤1 and x(t)=0, Otherwise	4	2	2
b)	List and discuss all the properties of Systems.	10	1	2
c)	Develop the signal z(n)=x(2n) y(n-4) with operations on signals x(n)=[3,2,1,0,1,2,3] and y(n)=[-1,-1,-1,-1,0,1,1,1,1]. The reference time index: x(n)=0 for n=0 (at center) and y(n)=0 for n=0 (at center)	6	2	3
3.a)	Prove that the convolution of any sequence with the unit sample sequence results in the same sequence: x(n)*δ(n)=x(n). OR x(n)*h(n)=x(n). if h(n)={1}	4	2	3
b)	Determine the output y(t) of a LTI system with impulse response h(t) with input x(t), using analytical method. $x(t) = \begin{cases} 1 \; for \; \lvert t \rvert \le 2 \\ 0 \; for \; \lvert t \rvert > 2 \end{cases}$ and $h(t) = u(t+1) - 2u(t) + u(t-1)$	8	2	3
c)	Determine the response of the system described by the difference equation, y(n)-(1/9) y(n-2) =x(n-1) with y (-1) =1, y (-2) =0 and x(n)=u(n).	8	2	3

OR

4.a)	Find the natural response of the system described by the differential equation, with initial conditions y(0)=3, (dy(t)/dt)=4 for t=0. $\frac{d^2 y(t)}{dt^2} + 3\frac{dy(t)}{dx(t)} + 2y(t) = 4e^{-3t}$ for t≥0.	7	2	3
b)	Evaluate and draw the signal y(n)=x(n)*h(n), where x(n)=1 for 0≤n≤4 and x(n)=0, otherwise, and h(n)=a^n for 0≤n≤6 and h(n)=0, otherwise, for a>1	7	1	2
c)	Draw the Direct form-I and Direct form-II implementation for the difference equation y(n)+(1/2) y(n-1)-y(n-3) =3x(n-1) +2x(n-2).	6	1	2
5.a)	State and Prove all the Properties of Fourier Transform (FT) .	10	3	2
b)	Find the DTFT of the following Signal **x(n)=a^n u(n)** .	5	4	3
c)	Find the FT of the Signal **x(t)=e^{-at} u(t)** and Plot its Magnitude and Phase Spectrum.	5	4	3

OR

6.a)	State and Prove all the Properties of Discrete Time Fourier Transform (DTFT)	10	3	3
b)	Find the FT of the Signal $x(t)=1$ for $-T \le t \le +T$ and Plot its magnitude and Phase Spectrum	5	4	3
c)	Find the DTFT of the Signal $x(n)=-a^n u(-n-1)$	5	4	3
7.a)	The impulse response of the continuous time system is given as, $h(t)=(1/RC)\, e^{-t/RC}\, u(t)$. Determine the frequency response and Plot the magnitude & Phase Plots.	10	4	3
b)	The system produces the output of $y(t)=e^{-t}\, u(t)$ for an input of $x(t)=e^{-2t}\, u(t)$. Determine the impulse response and frequency response of the system.	10	4	3

OR

8.a)	The impulse response of the continuous time system is given as, $h(t)=(1/RC)\, e^{-t/RC}\, u(t)$. Determine the frequency and Plot the magnitude & Phase Plots.	10	4	3
b)	The system produces the output of $y(t)=e^{-t}\, u(t)$ for an input of $x(t)=e^{-2t}\, u(t)$. Determine the impulse response and frequency response of the system.	10	4	3
9.a)	State and Prove all the properties of Z-Transform	10	2	2
b)	Solve the following difference equation using unilateral Z-Transform $y(n)-(3/2)\, y(n-1)+(1/2)\, y(n-2)=x(n)$ for $n \ge 0$ with initial condition $y(-1)=4$, $y(-2)=10$ and $x(n)=(1/4)^n u(n)$	10	4	3

OR

10.a)	Explain the properties of ROC of Z-Transform. Find the Z-transform of the following signals with RoC: $x(n)=a^n u(n)$ and $y(n)=7(1/3)^n u(n)-6(1/2)^n u(n)$	10	2	2
b)	Find the inverse Z-transform of the following using Partial fraction method. $Y(z)= (1-(1/2)\, z^{-1}) / (1+(3/4)\, z^{-1}+(1/8)\, z^{-2})$ for $\lvert z\rvert > (1/2)$	10	4	3

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